

Notes: Gao, Kim & Sevilir (JFE, 2025)

Private equity in the hospital industry

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1 Big picture

- **Question.** What happens to hospitals after private equity (PE) acquisition: closures, employment, efficiency, prices, and patient outcomes?
- **Setting.** U.S. hospital M&A from 2001 to 2018, comparing PE-acquired hospitals, non-PE for-profit acquisitions, and matched control hospitals.
- **Core result.** PE-acquired hospitals do not experience excess closures; they improve operating profitability in the short run while cutting employment and wage expenditures.
- **Labor mechanism.** Cuts are concentrated in administrative workers. Core medical workers temporarily fall but recover in the long run.
- **Patient outcomes.** Patient demographics and average inpatient prices do not significantly change; patient satisfaction falls; mortality and readmission rates do not deteriorate.
- **Takeaway.** PE improves efficiency mainly by reducing administrative costs, with mixed implications for patient experience.

1.1 Expanded interpretation of the paper’s contribution

- **The paper is not only about whether PE hospitals close.** A simple “asset-stripping” story would predict closures or large service reductions. The paper instead shows that survival is high and that the main operational margin is labor composition.
- **The paper’s central distinction is between cost cutting and quality erosion.** PE ownership raises profitability and lowers employment costs, but the key question is whether the cuts come from administrative overhead or from medical inputs that directly affect care.
- **The strongest mechanism is administrative efficiency.** The results repeatedly point to administrative workers and administrative wage costs as the durable margin of adjustment; core medical workers recover in the long run.
- **The patient evidence gives a two-level conclusion.** Hard clinical outcomes such as mortality and readmissions do not deteriorate, but softer experience outcomes such as communication, help, overall rating, and recommendation decline.
- **The paper therefore separates three outcomes that are often conflated:** financial efficiency, patient experience, and clinical quality. PE improves the first, worsens the second, and does not show statistically detectable harm to the third.

Interpretation. The paper’s main contribution is nuanced: PE acquisition appears to change the production function of hospitals, not simply by cutting everything, but by reallocating away from administrative labor while largely preserving core clinical capacity in the long run.

2 Data and measurement

2.1 Hospital M&A and operating data

The paper studies 1,218 hospital M&A deals from 2001–2018 and focuses on 281 for-profit-acquirer deals involving 610 unique target hospitals. Hospital operating data come from HCRIS. Patient demographics and prices come from HCCI claims; satisfaction comes from HCAHPS; mortality and readmission outcomes come from Hospital Compare.

Interpretation. HCRIS allows the paper to distinguish core medical labor from administrative labor, which is central to interpreting cost cutting.

2.2 Matching and event windows

Each target hospital is matched to a non-acquired control hospital using Mahalanobis distance, same Census Division, same metropolitan-area status, and variables including beds, CMI, payer mix, outpatient ratio, profitability, employment, core workers, and administrative workers. The pre-window is $[-4, -1]$, short-run post-window is $[0, 4]$, and long-run post-window is $[5, 8]$.

Interpretation. Matching creates a hospital-specific comparison group for event-time analysis.

2.3 How to interpret the key outcomes

- **Profitability variables** capture whether PE owners improve operating margins and asset productivity after acquisition.
- **Employment and wage variables** capture the cost-reduction channel. Because wages and employment are in logs, coefficients can be read approximately as percentage changes.
- **Core versus administrative workers** is the most important measurement split. Core workers include physicians, nurses, and pharmacists; administrative workers support finance, accounting, scheduling, and other overhead functions.
- **Patient demographics and prices** test whether profitability comes from treating different patients or charging more for inpatient care.
- **HCAHPS satisfaction measures** capture patients' experience with communication, responsiveness, overall rating, and recommendation.
- **Mortality and readmissions** capture hard clinical quality, but the available sample is smaller and the outcomes are measured in rolling CMS windows.

Interpretation. The paper's empirical credibility depends on measuring several margins at once. Profitability alone could reflect price increases, selection, or quality cuts; the paper's richer outcome set allows the authors to distinguish among these explanations.

3 Models and empirical specifications

3.1 Matched difference-in-differences

$$Y_{e,i,t} = \beta_1 PETarget_{e,i,t} + \beta_2 NonPETarget_{e,i,t} + \gamma' X_{i,t} + \alpha_i + \mu_e + \tau_t + \varepsilon_{e,i,t}. \quad (1)$$

Interpretation. β_1 measures post-acquisition changes at PE targets relative to matched controls. β_2 gives the corresponding effect for non-PE for-profit acquirers.

- If $\beta_1 > 0$ for profitability, the paper interprets this as improved operating performance at PE targets.
- If $\beta_1 < 0$ for employment or wages, the estimate reflects labor-cost reduction after PE acquisition.
- Comparing β_1 with β_2 asks whether PE ownership is different from ordinary for-profit acquisition, not merely different from no acquisition.

- Hospital fixed effects absorb persistent hospital quality and business-model differences; matched-pair fixed effects keep the comparison local to the matched design; event-time fixed effects absorb common acquisition-window shocks.

Interpretation. The coefficient is not simply a before-after change. It is a change relative to matched non-target hospitals and relative to non-PE acquisitions, which is why the tables can speak to what is distinctive about PE.

3.2 CPOM identification design

$$Y_{e,i,t} = \beta_1 CPOMRelaxation_{e,i,t} \times Post_{e,t} + \beta_2 CPOMTightening_{e,i,t} \times Post_{e,t} + \gamma' X_{i,t} + \alpha_i + \mu_e + \tau_{e,t} + \nu_{e,i,t}. \quad (2)$$

Interpretation. CPOM relaxations make PE entry easier by relaxing legal constraints on non-physician-owned corporations practicing medicine or employing physicians.

- The CPOM design is useful because it is tied to the legal feasibility of PE entry rather than the performance of a specific hospital.
- A relaxation of CPOM should raise PE acquisition probability if legal restrictions are binding.
- If CPOM relaxation also predicts lower administrative employment and higher profitability, it supports the view that PE entry causes operating restructuring.
- CPOM tightening is included as a falsification-like comparison: symmetric effects under tightening would be less consistent with a clean PE-entry mechanism.

Interpretation. This design does not replace the matched difference-in-differences analysis; it triangulates the mechanism with a different source of variation.

3.3 Mortality and readmission first differences

$$\Delta Y_{e,i,\tau} = \beta_1 PETarget_{e,i,\tau} + \beta_2 NonPETarget_{e,i,\tau} + \gamma' \Delta X_{i,t} + \mu_e + \nu_{e,i,\tau}. \quad (3)$$

Interpretation. Patient mortality and readmission tests compare changes from pre-acquisition windows to post-acquisition windows.

- The first-difference design is used because CMS mortality and readmission measures are reported with rolling windows.
- The estimates ask whether PE targets experience larger changes in clinical outcomes than comparable hospitals.
- The lack of significant deterioration should be read carefully: it means the paper does not detect worsening in these hard outcomes, not that quality cannot worsen on unmeasured dimensions.

Interpretation. This is why the paper’s patient-quality conclusion is split: no detected harm in hard outcomes, but clear deterioration in patient satisfaction.

4 Identification and empirical design

- Matching reduces observable differences between target and control hospitals.
- Hospital, matched-pair, and event-time fixed effects absorb persistent hospital differences, pair-level comparisons, and common event dynamics.
- Event studies check whether changes begin only after PE acquisition.
- CPOM legal changes provide additional variation in PE entry feasibility.
- Limitations include nonrandom PE selection, possible state-level correlates of CPOM changes, and smaller patient-outcome samples.

Interpretation. The empirical case is strongest because matching, event timing, CPOM variation, and labor heterogeneity point to the same administrative-cost-cutting channel.

4.1 Why the evidence hangs together

- **Survival evidence** rules against the strongest asset-stripping interpretation, because PE targets do not exhibit excess closure relative to matched controls.
- **Profitability evidence** shows an economically meaningful short-run improvement in hospital operations.
- **Labor evidence** identifies how profitability improves: employment and wage expenditures fall.
- **Worker-composition evidence** narrows the mechanism further: administrative workers remain lower while core workers recover.
- **CPOM evidence** connects the operating changes to legal variation that affects PE entry.
- **Patient-mix and price evidence** rules out two alternative mechanisms: attracting different patients or broadly increasing inpatient prices.
- **Patient satisfaction and clinical-outcome evidence** gives the welfare nuance: patients report worse experience, but mortality and readmission rates do not worsen in the observed data.

Interpretation. The paper’s causal argument is cumulative. No single table fully establishes the mechanism, but the pattern across tables is internally consistent.

4.2 Important caveats when interpreting the results

- The estimates are for hospitals acquired between 2001 and 2018 and may not fully describe more recent PE strategies or post-COVID hospital conditions.
- The price evidence focuses on inpatient prices in the HCCI claims sample; outpatient pricing could behave differently.
- Mortality and readmission outcomes are available for fewer hospitals and are measured for specific conditions, so they cannot cover every possible dimension of quality.
- Satisfaction declines may reflect administrative staff reductions, general disruption from acquisition, or changes in nonclinical services such as scheduling, billing, responsiveness, and amenities.
- The paper does not directly measure debt loads, dividend recapitalizations, or financial fragility after PE buyouts.

Interpretation. The correct reading is not “PE is harmless” or “PE is uniformly harmful.” The evidence supports a narrower conclusion: PE raises efficiency through administrative cost reductions, but this comes with lower patient experience and unresolved welfare trade-offs.

5 Tables and figures

5.1 Market and sample construction

Read: Table 1 compares unmatched targets and non-targets. Figure 1 shows M&A activity over time. Figure 2 shows the composition of PE and non-PE targets.

- PE targets are not randomly selected hospitals; target hospitals differ from never-acquired hospitals before matching.
- The deal composition figure shows why it is important to compare PE targets not only with controls, but also with non-PE for-profit acquisitions.
- The market activity figure motivates the question: PE ownership is large enough in the hospital sector to have meaningful labor and patient-care implications.

Economic message. Target hospitals differ from never-acquired hospitals before matching, and PE deals are economically important. The paper therefore needs a matched comparison design rather than a simple raw comparison.

Interpretation. These visuals establish the empirical setting. They do not yet identify treatment effects, but they show why selection and acquisition type must be handled carefully.

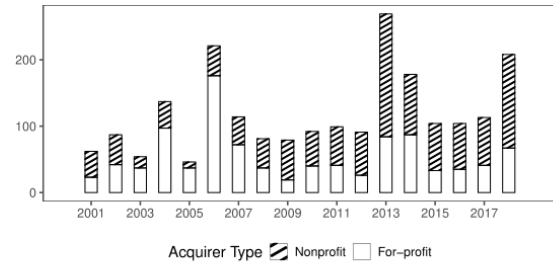
Table 1

Summary statistics for the initial (unmatched) sample.

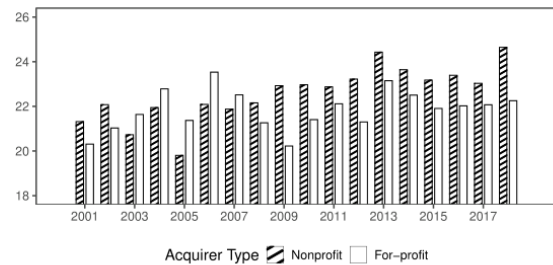
This table reports the summary statistics for the main variables used in our study. The sample includes target hospital observations during the four years prior to their acquisition and all observations from non-target hospitals. Target – Non Target represents the difference between the two groups. Detailed variable definitions are provided by [Appendix A](#). *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

	Non Target	Target	Target–Non Target
<i>Log(Employment)</i>	6.58	6.38	−0.20***
<i>Log(Core Workers)</i>	3.97	3.84	−0.12***
<i>Log(Admin Workers)</i>	4.39	4.11	−0.28***
<i>Log(Total Wages)</i>	17.36	17.10	−0.26***
<i>Beds</i>	142.24	168.18	25.94***
<i>CMI</i>	1.38	1.36	−0.02***
<i>%Medicare</i>	0.44	0.41	−0.03***
<i>%Medicaid</i>	0.13	0.14	0.02***
<i>%Outpatient</i>	0.55	0.41	−0.14***
Obs	49 230	1140	

(a) Table 1



(A) Number of Target Hospitals



(B) Total Asset Value of Target Hospitals (log)

(b) Figure 1

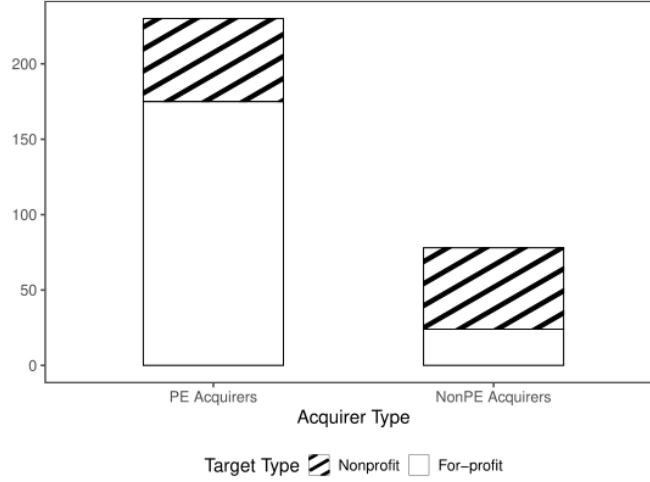


Fig. 2: Composition of target hospitals. This figure reports the number of target

Figure 2: target composition

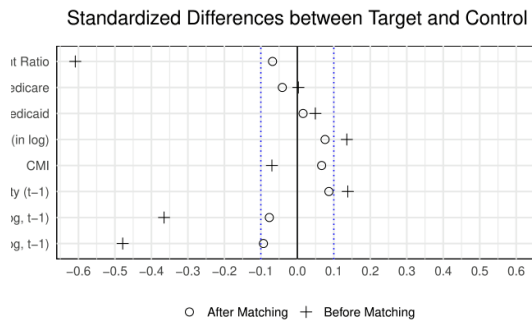
5.2 Matching and survival

Read: Figure 3 shows that standardized differences fall below 0.1 after matching. Figure 4 shows PE target survival remains high. Table 2 summarizes matched hospital, patient, and price variables.

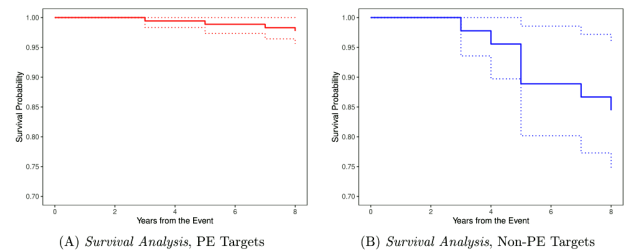
- The covariate-balance figure is central because PE targets and controls differ before matching.
- Survival rates overlap across PE targets, non-PE targets, and controls, and the paper reports only a small number of closures in the matched sample.
- Table 2 confirms that matched controls look similar across hospital operations, patient characteristics, and price-related variables.

Economic message. The matching design produces a balanced comparison group and weakens the simple claim that PE gains come from shutting hospitals down.

Interpretation. The survival result changes the interpretation of subsequent profitability gains. If PE targets remain open, then higher profitability likely comes from operating changes rather than closure and asset liquidation.



(a) Figure 3



Survival analysis of PE and non-PE targets. This figure shows the survival rates of target hospitals in each year after the acquisition. Panel A (B) reports the odds of target hospitals of PE (non-PE) acquirers. The solid lines represent the Kaplan-Meier estimates, while the dashed lines show their 95-percent confidence

(b) Figure 4

Table 2
Summary statistics for the matched sample.
This table reports the summary statistics for the matched sample of targets and controls. We track target and control hospitals during the $t-4$, $t+8$ event period, where t is defined by the acquisition of a hospital. Panel A reports hospital characteristics. Panel B presents summary statistics on patient satisfaction and outcomes. Panel C documents the summary statistics for risk-adjusted prices and patient demographics, obtained from the HCCI dataset. Detailed variable definitions are provided by [Appendix A](#).

(A) Hospital Characteristics						
	Obs	Mean	Std	Median	P25	P75
<i>Employment</i>	4,329	865.54	626.10	708.58	451.88	1083.62
<i>Log(Employment)</i>	4,329	6.52	0.72	6.56	6.12	6.99
<i>Core Workers</i>	4,328	67.14	62.70	47.86	25.32	89.51
<i>Log(Core Workers)</i>	4,328	3.86	0.88	3.89	3.27	4.51
<i>Admin Workers</i>	4,328	82.02	59.74	67.81	42.91	162.29
<i>Log(Admin Workers)</i>	4,328	4.21	0.66	4.23	3.78	4.64
<i>Total Wages (mil\$)</i>	4,344	48.21	37.89	38.60	21.76	62.13
<i>Log(Total Wages)</i>	4,344	17.39	0.83	17.47	16.90	17.94
<i>Beds</i>	4,399	181.09	121.64	148.00	101.00	235.00
<i>CMI</i>	4,309	1.38	0.21	1.38	1.25	1.54
<i>%Medicare</i>	4,399	0.38	0.13	0.38	0.29	0.47
<i>%Medicaid</i>	4,399	0.15	0.11	0.12	0.06	0.21
<i>%Outpatient</i>	4,398	0.42	0.15	0.40	0.30	0.53

(B) Patient Satisfaction and Patient Outcomes						
	Obs	Mean	Std	Median	P25	P75
<i>Mortality for Heart Attack (AMI)</i>	1,659	15.36	1.70	15.30	14.10	16.40
<i>Mortality for Heart Failure</i>	1,852	11.30	1.49	11.00	10.00	12.00
<i>Mortality for Pneumonia</i>	1,864	12.22	2.44	11.80	10.40	13.60
<i>Readmission for Heart Attack (AMI)</i>	1,244	18.90	1.70	19.00	17.70	20.00
<i>Readmission for Heart Failure</i>	1,566	23.91	2.07	24.00	22.30	25.40
<i>Readmission for Pneumonia</i>	1,579	17.93	1.59	18.00	16.90	19.00
<i>Nurse Comm.</i>	2,427	2.67	0.10	2.68	2.61	2.74
<i>Doctor Comm.</i>	2,427	2.73	0.08	2.73	2.68	2.78
<i>Receive Help</i>	2,427	2.47	0.13	2.48	2.38	2.56
<i>Hospital Rating</i>	2,427	2.53	0.13	2.54	2.45	2.61
<i>Recommendation</i>	2,427	2.58	0.12	2.59	2.51	2.67

(C) Hospital Prices and Patient Characteristics (Source: HCCI)						
	Obs	Mean	Std	Median	P25	P75
Price For:						
<i>Inpatient</i>	606	15 580.81	4064.60	15 162.60	13 079.04	17 361.56
<i>Hip Replacement</i>	425	24 909.85	9417.38	23 930.20	18 749.61	29 869.37
<i>Knee Replacement</i>	500	24 261.13	9311.91	22 598.13	17 790.12	29 782.37
<i>C-Section</i>	516	9510.26	3839.28	9145.81	6803.50	11 432.77
<i>Vaginal Delivery</i>	532	6103.88	2418.38	5936.74	4313.72	7431.08
<i>PTCA</i>	276	26 632.11	9535.16	24 799.47	19 545.93	32 864.82
<i>Colonoscopy</i>	673	1902.95	996.96	1723.64	1147.37	2448.31
<i>MRI</i>	619	1121.82	568.70	1035.83	648.95	1501.72
Log(Price) For:						
<i>Inpatient</i>	606	9.62	0.24	9.63	9.48	9.76
<i>Hip Replacement</i>	424	10.05	0.43	10.09	9.84	10.31
<i>Knee Replacement</i>	500	10.02	0.42	10.03	9.79	10.30
<i>C-Section</i>	516	9.00	1.98	9.12	8.83	9.34
<i>Vaginal Delivery</i>	530	8.64	0.40	8.69	8.37	8.91
<i>PTCA</i>	276	10.12	0.37	10.12	9.88	10.40
<i>Colonoscopy</i>	671	7.41	0.57	7.45	7.05	7.81
<i>MRI</i>	618	6.89	0.54	6.95	6.48	7.31
Patient Characteristics:						
<i>Female (Simple Average)</i>	769	0.59	0.05	0.59	0.57	0.63
<i>Female (Weighted Average)</i>	769	0.57	0.07	0.57	0.54	0.60
<i>Age (Simple Average)</i>	769	4.06	0.40	4.11	3.91	4.29
<i>Age (Weighted Average)</i>	769	4.44	0.47	4.50	4.29	4.71
<i>Income (Simple Average)</i>	769	61 514.45	19 472.13	55 949.80	48 291.93	68 470.64
<i>Income (Weighted Average)</i>	741	59 871.64	18 392.87	54 685.72	47 455.93	66 483.43
<i>Log(Income (Simple Average))</i>	769	10.99	0.28	10.93	10.79	11.13
<i>Log(Income (Weighted Average))</i>	741	10.96	0.27	10.91	10.77	11.10

Table 2

5.3 Main operating effects

Read: Table 3 shows positive short-run profitability effects. Table 4 shows PE targets cut employment and wages. Table 5 shows administrative-worker cuts persist while core-worker cuts reverse. Figure 5 shows the dynamic pattern.

- PE targets increase operating income and ROA by about four percentage points in the first four years after acquisition.
- Total employment falls by roughly six percent and wage expenditures fall by roughly seven percent in the short run; wage costs remain about nine percent lower in the long run.
- Core medical workers fall temporarily by about sixteen percent but recover by years five to eight.
- Administrative workers fall by about seventeen percent initially and by more than twenty percent in the longer window.
- The dynamic figure is consistent with restructuring: core medical capacity is disrupted early, while administrative rationalization persists.

Economic message. PE improves efficiency by cutting administrative costs rather than permanently reducing core medical labor.

Interpretation. This is the main mechanism table group. The paper’s interpretation depends heavily on the distinction between temporary core-worker disruption and persistent administrative-worker reduction.

Without that distinction, the employment cuts could be read as quality deterioration; with it, they look more like overhead reduction.

Table 3

Profitability at target hospitals.

This table examines changes in profitability at target hospitals around acquisitions. The dependent variable for columns (1) and (2) is *Gross Margin*, which is net income from service to patients (as given in HCRIS) over net patient revenues (as given in HCRIS). The dependent variable for columns (3) and (4) is *OI/TA*, which is net income from service to patients (as given in HCRIS) over total assets. The dependent variable for column (5) and (6) is *ROA*, which is net income (total income-total other expenses, as given in HCRIS) over total assets. *PE Target* turns to one after a hospital is acquired by a PE acquirer. *Non-PE Target* turns to one after a hospital is acquired by a non-PE acquirer. Rows with H_{it} 's provide p -values from Wald Chi-square tests indicating whether two coefficients are statistically significantly different from each other. Hospital Controls include the log of total beds (*Log(Beds)*), case mix index (*CMJ*), percentage of patients with Medicare (*%Medicare*), percentage of patients with Medicaid (*%Medicaid*), and the percentage of patients that are outpatients (*%Outpatient*). County Controls include the percentage of Black residents (*%Black*), the percentage of Asian residents (*%Asian*), log of population (*Log(Pop)*), and the log of one bedroom rent in a county (*Log(FMR)*). See [Appendix A](#) for variable definitions. t -statistics are reported in parentheses and standard errors are heteroskedasticity robust and clustered by hospital. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Dep. Var.:	<i>Gross Margin</i>		<i>OI/TA</i>		<i>ROA</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
Post-Event Window	[0, 4]	[5, 8]	[0, 4]	[5, 8]	[0, 4]	[5, 8]
<i>PE Target</i>	0.0015 (0.14)	0.0214 (0.86)	0.0399** (2.36)	0.0333 (0.88)	0.0399*** (2.65)	0.0343 (1.09)
<i>NonPE Target</i>	0.0206 (1.00)	0.0321 (0.79)	0.0465 (1.23)	0.0021 (0.04)	0.0013 (0.05)	-0.0133 (-0.35)
Hospital Controls	Yes	Yes	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	Yes	Yes	Yes
Hospital FEs	Yes	Yes	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes	Yes	Yes
Event Time FEs	Yes	Yes	Yes	Yes	Yes	Yes
H_{it} : PE = NonPE	0.36	0.81	0.87	0.61	0.15	0.30
Obs	3,685	2,230	3,677	2,225	3,677	2,225
Adj. R^2	0.58	0.56	0.56	0.55	0.54	0.50

(a) Table 3

Table 4

Employment and wage expenditures at target hospitals.

This table examines the changes in employment and wages at target hospitals around acquisitions. The dependent variable in column (1) and (2) is the log of total employees (measured in full-time equivalent employees based on employed hours). The dependent variable in column (3) and (4) is the log of total wages. *PE Target* turns to one after a hospital is acquired by a PE acquirer. *NonPE Target* turns to one after a hospital is acquired by a non-PE acquirer. Rows with H_{it} 's provide p -values from Wald Chi-square tests indicating whether two coefficients are statistically significantly different from each other. Hospital Controls and County Controls are defined in the same way as in [Table 3](#). See [Appendix A](#) for variable definitions. t -statistics are reported in parentheses and standard errors are heteroskedasticity robust and clustered by hospital. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Dep. Var.:	<i>Log(Employment)</i>		<i>Log(Total Wages)</i>	
	(1)	(2)	(3)	(4)
Post-Event Window	[0, 4]	[5, 8]	[0, 4]	[5, 8]
<i>PE Target</i>	-0.0612*** (-4.08)	-0.0630* (-1.80)	-0.0721*** (-3.67)	-0.0855** (-2.01)
<i>NonPE Target</i>	-0.0259 (-0.91)	-0.0282 (-0.54)	-0.0604* (-1.93)	-0.0489 (-0.83)
Hospital Controls	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	Yes
Hospital FEs	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes
Event Time FEs	Yes	Yes	Yes	Yes
H_{it} : PE = NonPE	0.23	0.52	0.71	0.54
Obs	3,691	2,231	3,691	2,231
Adj. R^2	0.98	0.98	0.98	0.98

(b) Table 4

Table 5

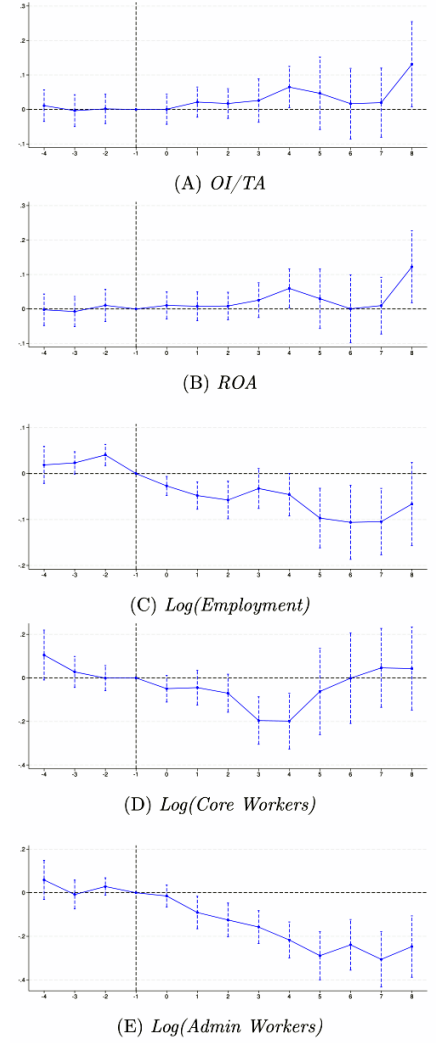
Core and administrative workers at target hospitals.

This table examines the changes in core workers and administrative workers at target hospitals around acquisitions. Panel A reports the results for the number of core and administrative workers. The dependent variable for columns (1) and (2) is the log of total number of core workers, i.e., Log(Core Workers) . The dependent variable for columns (3) and (4) is the log of total number of administrative workers, i.e., $\text{Log(Admin Workers)}$. Panel B reports the results contrasting short-run and long-run changes in core workers and administrative workers at target hospitals. The dependent variable for column (1) is Log(Core Workers) , while that for column (2) is $\text{Log(Admin Workers)}$. Long-run turns to one for year [5, 8] following the acquisition event. PE Target turns to one after a hospital is acquired by a PE acquirer. NonPE Target turns to one after a hospital is acquired by a non-PE acquirer. Rows with H_0 's provide p -values from Wald Chi-square tests indicating whether two coefficients are statistically significantly different from each other. Hospital Controls and County Controls are defined in the same way as in Table 3. See Appendix A for variable definitions. t -statistics are reported in parentheses and standard errors are heteroskedasticity robust and clustered by hospital. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

(A) Log Number of Core and Administrative Workers				
Dep. Var.:	Log(Core Workers)		$\text{Log(Admin Workers)}$	
	(1)	(2)	(3)	(4)
Post-Event Window	[0, 4]	[5, 8]	[0, 4]	[5, 8]
<i>PE Target</i>	-0.1620*** (-4.74)	-0.0120 (-0.16)	-0.1656*** (-5.56)	-0.2027*** (-3.31)
<i>NonPE Target</i>	-0.2868*** (-4.49)	-0.3598** (-2.20)	-0.0310 (-0.59)	-0.0314 (-0.23)
Hospital Controls	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	Yes
Hospital FEs	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes
Event Time FEs	Yes	Yes	Yes	Yes
$H_0: \text{PE} = \text{NonPE}$	0.05	0.05	0.01	0.21
Obs	3,691	2,231	3,691	2,231
Adj. R^2	0.91	0.90	0.90	0.90

(B) Long-run versus Short-run Changes		
Dep. Var.:	Log(Core Workers)	$\text{Log(Admin Workers)}$
	(1)	(2)
Post-Event Window	[0, 8]	[0, 8]
<i>PE Target</i>	-0.1598*** (-4.75)	-0.1663*** (-5.65)
<i>PE Target</i> $\times$ <i>Long-run</i>	0.0943* (1.82)	-0.0649* (-1.82)
<i>NonPE Target</i>	-0.2938*** (-4.69)	-0.0330 (-0.64)
<i>NonPE Target</i> $\times$ <i>Long-run</i>	-0.0666 (-0.62)	-0.0355 (-0.48)
Hospital Controls	Yes	Yes
County Controls	Yes	Yes
Hospital FEs	Yes	Yes
Matched Pair FEs	Yes	Yes
Event Time FEs	Yes	Yes
Obs	4,281	4,281
Adj. R^2	0.90	0.90

(a) Table 5



5. Dynamic effect of PE acquirers. This figure shows the changes in profitability, employment, and worker composition for PE-acquired hospitals relative to the matched control hospitals over the [-4, +8]-year event window. All regressions include a full set of fixed effects and controls as in the baseline analysis, and further include zone-by-year interactive fixed effects to help remove potential confounders from local conditions. In each panel, the dots and intervals represent the

(b) Figure 5

5.4 CPOM and nonprofit heterogeneity

Read: Table 6 shows CPOM relaxation increases PE acquisition likelihood and is associated with higher ROA and fewer administrative workers. Table 7 shows nonprofit targets experience especially large administrative-worker cuts.

- CPOM relaxation raises the probability of PE acquisition, consistent with the legal channel affecting PE entry.
- Following CPOM relaxation, ROA increases and administrative-worker counts fall, while core-worker changes are not the central margin.
- Nonprofit targets have larger administrative-worker reductions than for-profit targets, consistent with nonprofit hospitals having more administrative slack or different objective functions before acquisition.
- The nonprofit heterogeneity result helps explain why PE can create operating gains without necessarily

changing patient prices or patient mix.

Economic message. CPOM evidence supports the PE mechanism; nonprofit targets offer the largest administrative-efficiency gains.

Interpretation. These results make the mechanism more credible because they show that the largest PE effects occur where the model of administrative restructuring predicts they should occur: after legal barriers fall and in hospitals with likely pre-existing administrative slack.

Table 6

CPOM analysis.

This table examines the changes in profitability, employment, and core and administrative workers at hospitals around CPOM reforms at the state level. The analysis uses a matched sample of hospitals, where each hospital in treated states (i.e., states that changed their CPOM doctrines) is matched with a hospital in control states (i.e., states without any CPOM events during our sample period). We analyze changes at hospitals around the [-4,+8] CPOM event window. In Panel A, we examine how the relaxation and tightening of CPOM enforcement at the state level affect the likelihood that a hospital is acquired by PE investors. In Panel B, we examine changes in hospital profitability, employment, and core and administrative workers around CPOM events. *CPOM Relaxation* is an indicator for states that experienced a relaxation in CPOM doctrines. *CPOM Tightening* is an indicator that turns to one for states that experienced a tightening in CPOM doctrines. *Post* indicates years after the relaxation or tightening event. *PE Target* turns to one after a hospital is acquired by a PE acquirer. *Hospital Controls* and *County Controls* are defined in the same way as in Table 3. See Appendix A for variable definitions. *t*-statistics are reported in parentheses and standard errors are heteroskedasticity robust and clustered by state. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

(A) The Effects of CPOM Events on PE Acquisition					
Dep. Var.: PE Target	(1)				
CPOM Relaxation × Post	0.0914** (2.09)				
CPOM Tightening × Post	0.0096 (0.79)				
Hospital Controls	Yes				
County Controls	Yes				
Hospital FEs	Yes				
Matched Pair FEs	Yes				
Event Time FEs	Yes				
Obs	5,287				
Adj. R ²	0.57				
(B) The Effects of CPOM Events on Profitability and Employment					
Dep. Var.:	OI/AT (2)	ROA (3)	Log(Total Employment) (4)	Log(Core Workers) (5)	Log(Admin Workers) (5)
CPOM Relaxation × Post	0.0108 (1.12)	0.0205* (1.85)	-0.0431** (-2.28)	0.0267 (0.91)	-0.0916** (-2.56)
CPOM Tightening × Post	0.0134 (1.14)	0.0069 (0.81)	0.0376 (1.66)	0.0571 (1.56)	0.0171 (0.47)
Hospital Controls	Yes	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	Yes	Yes
Hospital FEs	Yes	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes	Yes
Event Time FEs	Yes	Yes	Yes	Yes	Yes
Obs	5,278	5,278	5,287	5,286	5,282
Adj. R ²	0.45	0.36	0.99	0.95	0.92

Table 7

Nonprofit vs. for-profit targets.

This table presents heterogeneity of our results across previously nonprofit and for-profit targets. We report the results for *OI/AT* (column (1) and (2)), *ROA* (column (3) and (4)), the log of total employment (column (5) and (6)), the log of core medical workers (column (7) and (8)), and the log of administrative workers (column (9) and (10)) at target hospitals around acquisitions. Target hospitals' for-profit status is characterized based on its status prior to the acquisition. All regressions include the indicator of non-PE acquisition, but we suppress the coefficients for brevity. Rows with *H*₀'s provide *p*-values from Wald Chi-square tests indicating whether two coefficients are statistically significantly different from each other. *Hospital Controls* and *County Controls* are defined in the same way as in Table 3. See Appendix A for variable definitions. *t*-statistics are reported in parentheses and standard errors are heteroskedasticity robust and clustered by hospital. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Dep. Var.:	<i>OI/AT</i>		<i>ROA</i>		<i>Log(Employment)</i>		<i>Log(Core Workers)</i>		<i>Log(Admin Workers)</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Post-Event Window	[5, 4]	[5, 8]	[5, 8]	[5, 8]	[5, 8]	[5, 8]	[5, 8]	[5, 8]	[5, 8]	[5, 8]
<i>PE Target</i> × <i>Nonprofit Target</i>	0.0627** (2.33)	0.0489 (0.86)	0.0320 (1.17)	0.0461 (0.90)	-0.1259*** (-4.43)	-0.0902* (-1.77)	-0.1399** (-2.40)	-0.0080 (-0.07)	-0.2214*** (-4.07)	-0.1346*** (-4.07)
<i>PE Target</i> × <i>For-profit Target</i>	0.0336* (1.82)	0.0213 (0.48)	0.0421*** (2.61)	0.0251 (0.71)	-0.0431*** (-2.82)	-0.0420 (-1.04)	-0.1681*** (-4.62)	-0.0151 (-0.17)	-0.1501*** (-4.83)	-0.1604 (-1.33)
Hospital Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hospital FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Event Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>H</i> ₀ : For-Profit = Nonprofit	0.30	0.68	0.73	0.70	0.00	0.41	0.63	0.96	0.19	0.01
Obs	3,677	2,225	3,677	2,225	3,691	2,221	3,691	2,231	3,691	2,231
Adj. R ²	0.56	0.55	0.54	0.50	0.98	0.98	0.91	0.90	0.90	0.90

5.5 Operations, patient mix, and prices

Read: Table 8 shows administrative wage rates fall in the long run. Table 9 shows no large operating-scale expansion. Table 10 shows no significant patient-demographic shift. Table 11 and Figure 7 show no broad inpatient price increase.

- Core wage rates do not materially change at PE targets, which supports the interpretation that PE does not permanently degrade core clinical labor markets.
- Administrative wage rates decline, reinforcing the administrative-cost channel.
- Hospital beds, case mix, revenues, and operating scale do not show a broad expansion story.
- Patient demographics do not shift significantly, so the results are not driven by serving systematically different patients.
- Inpatient prices do not rise broadly, which distinguishes the paper from concerns that PE gains are mainly achieved through higher hospital prices.

Economic message. PE profitability gains do not appear to come from shifting patient mix or broadly raising inpatient prices.

Interpretation. This table group rules out several alternative explanations. The paper is not primarily finding revenue extraction through patient selection or inpatient price increases; it is finding cost-side restructuring.

Table 8

Wage rates for core and administrative workers at target hospitals.

This table examines the changes in per hour salary paid to core workers and administrative workers at target hospitals around acquisitions. In columns (1) and (2), we present results related to *Log(Core Wage Rate)*, the log of hourly wage rate for core workers. In columns (3) and (4), we present results related to *Log(Admin Wage Rate)*, the log of hourly wage rate for administrative workers. *PE Target* turns to one after a hospital is acquired by a PE acquirer. *NonPE Target* turns to one after a hospital is acquired by a non-PE acquirer. Rows with H_0 's provide p -values from Wald Chi-square tests indicating whether two coefficients are statistically significantly different from each other. Hospital Controls and County Controls are defined in the same way as in Table 3. See Appendix A for variable definitions. t -statistics are reported in parentheses and standard errors are heteroskedasticity robust and clustered by hospital. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Dep. Var.:	<i>Log(Core Wage Rate)</i>		<i>Log(Admin Wage Rate)</i>	
	(1)	(2)	(3)	(4)
Post-Event Window	[0, 4]	[5, 8]	[0, 4]	[5, 8]
<i>PE Target</i>	-0.0049 (-0.33)	-0.0455 (-1.41)	-0.0196 (-1.29)	-0.0708** (-2.35)
<i>NonPE Target</i>	-0.0531* (-1.83)	0.0139 (0.29)	-0.0414* (-1.68)	0.0253 (0.54)
Hospital Controls	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	Yes
Hospital FEs	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes
Event Time FEs	Yes	Yes	Yes	Yes
H_0 : PE = NonPE	0.10	0.24	0.38	0.05
Obs	3,643	2,203	3,581	2,174
Adj. R^2	0.71	0.71	0.81	0.79

(a) Table 8

Table 9

Size and operating characteristics at target hospitals.

This table examines changes in size and operating characteristics at target hospitals around acquisitions. Panel A examines hospitals' operating scale, measured by the log number of beds, patients, and sales. Panel B examines the composition of hospitals patients or hospital operations, measured by CMI, outpatient ratio, and the percentage of Medicare and Medicaid patients. *PE Target* turns to one after a hospital is acquired by a PE acquirer. *NonPE Target* turns to one after a hospital is acquired by a non-PE acquirer. Rows with H_0 's provide p -values from Wald Chi-square tests indicating whether the two coefficients are statistically significantly different from each other. Control variables are the same as in Table 3. See Appendix A for variable definitions. t -statistics are reported in parentheses and standard errors are heteroskedasticity robust and clustered by hospital. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

(A) Hospital Operating Scale								
Dep. Var.:	<i>Log(Beds)</i>		<i>Log(Patients)</i>		<i>Log(Gross Patient Sales)</i>		<i>Log(Net Patient Sales)</i>	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post-Event Window	[0, 4]	[5, 8]	[0, 4]	[5, 8]	[0, 4]	[5, 8]	[0, 4]	[5, 8]
<i>PE Target</i>	-0.0068 (-0.40)	0.0225 (0.45)	-0.0215 (-1.06)	-0.1008* (-1.69)	-0.0149 (-0.61)	-0.0619 (-0.93)	-0.0685*** (-3.00)	-0.0173 (-0.27)
<i>NonPE Target</i>	-0.0588** (-2.33)	-0.0011 (-0.02)	-0.1033** (-2.53)	-0.1051 (-1.42)	-0.0880** (-2.26)	-0.0617 (-0.76)	-0.1100** (-2.59)	-0.0312 (-0.39)
Hospital Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hospital FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Event Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
H_0 : PE = NonPE	0.04	0.68	0.04	0.96	0.07	1.00	0.27	0.88
Obs	3,771	2,270	3,766	2,269	3,765	2,269	3,765	2,269
Adj. R^2	0.97	0.96	0.96	0.96	0.98	0.97	0.97	0.97
(B) Hospital Operation and Patient Composition								
Dep. Var.:	CMI		Outpatient Ratio		%Medicare		%Medicaid	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post-Event Window	[0, 4]	[5, 8]	[0, 4]	[5, 8]	[0, 4]	[5, 8]	[0, 4]	[5, 8]
<i>PE Target</i>	-0.0026 (-0.30)	0.0167 (0.78)	-0.0159*** (-3.41)	-0.0344** (-2.80)	-0.0125** (-2.18)	0.0062 (0.43)	-0.0116* (-1.83)	-0.0134 (-0.73)
<i>NonPE Target</i>	-0.0244* (-1.82)	-0.0167 (-0.62)	-0.0122 (-1.51)	-0.0304** (-2.27)	0.0116 (1.42)	0.0305* (1.75)	-0.0064 (-0.72)	-0.0319 (-1.51)
Hospital Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hospital FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Event Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
H_0 : PE = NonPE	0.10	0.27	0.64	0.77	0.00	0.18	0.58	0.42
Obs	3,705	2,240	3,770	2,269	3,771	2,270	3,771	2,270
Adj. R^2	0.91	0.89	0.95	0.94	0.89	0.88	0.79	0.75

(b) Table 9

Table 10

Patient demographics at target hospitals.

This table examines the changes in patient demographics at target hospitals around the [-4,+8] acquisition event window. Columns (1) and (2) examine changes in the proportion of female patients, while columns (3) and (4) examine the average age of patients. In columns (5) and (6), we investigate the average income of patients, as determined by the income level of their residential area. When aggregating patient information at the hospital level, we use either a simple average or a weighted average, with the weights being the total amount paid to the hospital for the service delivered to each patient during the year. *PE Target* turns to one after a hospital is acquired by a PE acquirer. *NonPE Target* turns to one after a hospital is acquired by a non-PE acquirer. Rows with H_0 's provide p -values from Wald Chi-square tests indicating whether two coefficients are statistically significantly different from each other. Control variables are the same as in Table 3. See Appendix A for variable definitions. t -statistics are reported in parentheses and standard errors are heteroskedasticity robust and clustered by hospital. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Dep. Var.:	Female		Age		Income (IRS, in log)	
	Simple	Weighted	Simple	Weighted	Simple	Weighted
	(1)	(2)	(3)	(4)	(5)	(6)
<i>PE Target</i>	-0.001 (-0.12)	-0.010 (-0.56)	-0.078 (-1.56)	-0.013 (-0.28)	-0.014 (-1.40)	0.003 (0.21)
<i>NonPE Target</i>	-0.003 (-0.30)	0.001 (0.06)	-0.055 (-1.12)	-0.078 (-1.28)	-0.006 (-0.55)	-0.006 (-0.40)
Hospital Controls	Yes	Yes	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	Yes	Yes	Yes
Hospital FEs	Yes	Yes	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes	Yes	Yes
Event Time FEs	Yes	Yes	Yes	Yes	Yes	Yes
H_0 : PE = NonPE	0.81	0.59	0.67	0.67	0.37	0.50
Obs	752	752	752	752	752	724
Adj. R^2	0.81	0.73	0.91	0.88	0.98	0.97

(a) Table 10

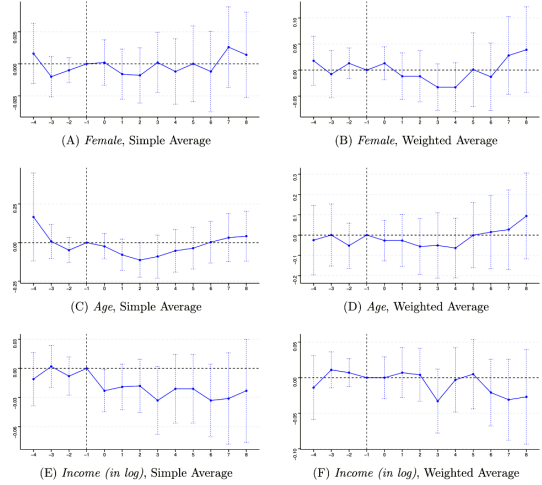


Fig. 6. Dynamic changes in patient demographics around PE acquisition. This figure shows the changes in patient demographics for PE-acquired hospitals relative to matched control hospitals over the [-4, +8]-year event window. The left (right) column presents the changes in the simple (weighted) average of patient characteristics.

(b) Figure 6

Table 11

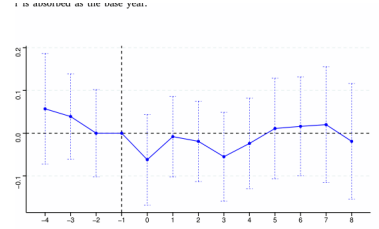
Inpatient and procedure prices at target hospitals.

This table examines the changes in prices charged by target hospitals for overall inpatient procedures and seven specific procedures around the [-4,+8] acquisition event window. In Panel A, we analyze inpatient price changes. In Panel B, we analyze price changes for seven procedures, namely, hip replacement, knee replacement, C-section, vaginal delivery, PTCA, colonoscopy, and MRI. The dependent variable is the log of prices. *PE Target* turns to one after a hospital is acquired by a PE acquirer. *NonPE Target* turns to one after a hospital is acquired by a non-PE acquirer. Rows with H_0 's provide p -values from Wald Chi-square tests indicating whether two coefficients are statistically significantly different from each other. Control variables are the same as in Table 3. See Appendix A for variable definitions. t -statistics are reported in parentheses and standard errors are heteroskedasticity robust and clustered by hospital. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

(A) Inpatient Prices (in log)							
Dep. Var.: Inpatient	(1)		(2)		(3)		
<i>PE Target</i>	-0.042 (-1.05)		-0.049 (-1.32)		-0.048 (-1.20)		
<i>NonPE Target</i>	0.012 (0.18)		0.013 (0.21)		0.022 (0.34)		
Hospital Controls			Yes		Yes		
County Controls					Yes		
Hospital FEs	Yes		Yes		Yes		
Matched Pair FEs	Yes		Yes		Yes		
Event Time FEs	Yes		Yes		Yes		
H_0 : <i>PE</i> = <i>NonPE</i>	0.40		0.30		0.27		
Obs	606		605		605		
Adj. R^2	0.84		0.84		0.84		

(B) Procedure Prices (in log)							
Dep. Var.:	Hip Replacement	Knee Replacement	C-Section	Vaginal Delivery	PTCA	Colonoscopy	MRI
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>PE Target</i>	0.128 (1.28)	0.014 (0.22)	-0.121 (-1.03)	-0.010 (-0.13)	0.029 (0.20)	0.294*** (2.60)	0.101 (0.76)
<i>NonPE Target</i>	0.306 (1.00)	-0.064 (-0.41)	-0.050 (-0.34)	-0.143 (-1.62)	0.351* (1.70)	0.150 (1.07)	0.052 (0.45)
Hospital Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hospital FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Event Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
H_0 : <i>PE</i> = <i>NonPE</i>	0.56	0.57	0.70	0.17	0.17	0.30	0.68
Obs	424	499	515	528	275	662	607
Adj. R^2	0.52	0.68	0.15	0.74	0.50	0.63	0.76

(a) Table 11



g. 7. Dynamic changes in hospital inpatient prices around PE acquisition. This figure shows the changes in the average inpatient prices at PE-acquired hospitals relative to their matched control hospitals over the [-4, +8]-year event window. We control for hospital-level and county-level factors as defined in Table 3, and include hospital fixed effects, matched-pair fixed effects, and event time fixed effects. The dots

(b) Figure 7

5.6 Patient satisfaction and clinical outcomes

Read: Table 12 shows patient satisfaction declines across many HCAHPS dimensions. Table 13 finds no significant deterioration in mortality or readmission outcomes at PE targets.

- Satisfaction falls for communication with nurses, communication with doctors, receiving help, overall rating, and willingness to recommend.
- The decline is present both in short-run and long-run windows for several dimensions, which suggests more than a one-time acquisition disruption.
- Mortality and readmission regressions do not show statistically significant deterioration for PE targets across heart attack, heart failure, and pneumonia outcomes.
- This creates a sharp distinction between patient experience and clinical outcomes.

Economic message. PE ownership appears to preserve hard clinical outcomes but reduce patient experience, plausibly because administrative support functions are reduced.

Interpretation. This is the key welfare nuance. Administrative reductions may not show up in mortality or readmissions, but they can still make hospitals feel less responsive, less communicative, or harder for patients to navigate.

6 Interpretive synthesis

- **Efficiency channel.** PE ownership raises operating profitability mainly by reducing labor costs, especially administrative labor.
- **Quality channel.** The paper separates quality into patient experience and clinical outcomes. Experience worsens; mortality and readmissions do not show detectable deterioration.
- **Selection channel.** Matching and covariate balance reduce concerns that all findings are driven by pre-existing hospital differences, though selection cannot be fully eliminated.

Table 12

Patient satisfaction at target hospitals.

This table examines the changes in patient satisfaction at target hospitals around acquisitions. In columns (1) and (2), we analyze patient satisfaction with nurses' communication, while in columns (3) and (4), we focus on satisfaction with doctors' communication. Columns (5) and (6) report patient satisfaction concerning whether patients received help as soon as they wanted, and Columns (7) and (8) report patients' overall rating of the hospital. In columns (9) and (10), we investigate patients' willingness to recommend the hospital to friends and family. *PE Target* turns to one after a hospital is acquired by a PE acquirer. *NonPE Target* turns to one after a hospital is acquired by a non-PE acquirer. Rows with H_0 's provide p -values from Wald Chi-square tests indicating whether two coefficients are statistically significantly different from each other. Control variables are the same as in Table 3. See Appendix A for variable definitions. t -statistics are reported in parentheses and standard errors are heteroskedasticity robust and clustered by hospital. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

Dep. Var.:	Nurse Comm.		Doctor Comm.		Receive Help		Hospital Rating		Recommendation	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Post-Event Window	[8.4]	[5.8]	[5.8]	[5.8]	[8.4]	[5.8]	[8.4]	[5.8]	[8.4]	[5.8]
<i>PE Target</i>	-0.0208*** (-3.08)	-0.0450*** (-2.67)	-0.0130* (-1.89)	-0.0268** (-1.86)	-0.0358*** (-2.42)	-0.0618*** (-1.58)	-0.0207 (-2.42)	-0.0714** (-2.40)	-0.0322** (-2.13)	-0.0544*** (-2.09)
<i>NonPE Target</i>	-0.0446*** (-3.51)	-0.0465 (-1.43)	-0.0211** (-2.08)	-0.0208 (-1.43)	-0.0517*** (-2.46)	-0.0405 (-1.00)	-0.0507*** (-2.54)	-0.0582 (-1.35)	-0.0488*** (-3.12)	-0.0548 (-1.40)
Hospital Controls										
County Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Hospital FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Event Time FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
H_0 : PE = NonPE	0.28	0.96	0.45	0.80	0.47	0.59	0.26	0.76	0.37	0.99
Obs	1,833	934	1,833	934	1,833	934	1,833	934	1,833	934
Adj. R ²	0.81	0.79	0.76	0.78	0.80	0.76	0.75	0.74	0.80	0.80

Table 13

Mortality and readmission rates at target hospitals.

This table examines the mortality and readmission rates at target hospitals around acquisitions. Panel A reports the results for mortality rates. The dependent variables are the 30-day risk-standardized mortality rate following heart attack hospitalization, heart failure hospitalization, and pneumonia hospitalization. Panel B reports the results for readmission rates. The dependent variables are the 30-day risk-standardized readmission rates for patients discharged from the hospital with a principal diagnosis of heart attack, heart failure, and pneumonia, respectively. Mortality rates and readmission rates are presented in percentage points. The regressions take a first-difference approach, with both the dependent variables and continuous control variables representing changes from the pre-acquisition window to a post-acquisition window. *PE Target* turns to one after a hospital is acquired by a PE acquirer. *NonPE Target* turns to one after a hospital is acquired by a non-PE acquirer. Rows with H_0 's provide p -values from Wald Chi-square tests indicating whether two coefficients are statistically significantly different from each other. Control variables are the same as in Table 3. See Appendix A for variable definitions. t -statistics are reported in parentheses and standard errors are heteroskedasticity robust and clustered by hospital. *, **, and *** indicate statistical significance at the 10%, 5%, and 1%, respectively.

(A) Changes in Mortality						
Dep. Var.:	Heart Attack (AMI)		Heart Failure		Pneumonia	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>PE Target</i>	-0.0702 (-0.14)	0.4779 (0.99)	0.2126 (0.36)	-0.5163 (-0.87)	0.6729 (1.29)	-0.1525 (-0.33)
<i>NonPE Target</i>	-1.2259 (-1.87)	-2.1883*** (-4.80)	0.8621 (1.47)	0.5946 (1.12)	0.6779 (0.79)	1.0721* (1.78)
Hospital Controls (differenced)	Yes	Yes	Yes	Yes	Yes	Yes
County Controls (differenced)	Yes	Yes	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes	Yes	Yes
H_0 : PE = NonPE	0.11	0.00	0.27	0.22	0.99	0.14
Obs	175	175	199	199	199	199
Adj. R ²	0.10	0.55	0.06	0.53	0.07	0.57
(B) Changes in Readmission						
Dep. Var.:	Heart Attack (AMI)		Heart Failure		Pneumonia	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>PE Target</i>	0.3644 (0.65)	-0.0570 (-0.11)	0.4897 (0.77)	0.2870 (0.54)	-0.1837 (-0.39)	-0.7147 (-1.46)
<i>NonPE Target</i>	-0.3770 (-0.58)	-0.3371 (-0.95)	0.6336 (0.89)	0.0371 (0.06)	-0.5383 (-0.92)	-0.8393** (-2.03)
Hospital Controls (differenced)	Yes	Yes	Yes	Yes	Yes	Yes
County Controls (differenced)	Yes	Yes	Yes	Yes	Yes	Yes
Matched Pair FEs	Yes	Yes	Yes	Yes	Yes	Yes
H_0 : PE = NonPE	0.20	0.63	0.85	0.75	0.55	0.86
Obs	126	126	163	163	163	163
Adj. R ²	0.13	0.57	0.18	0.57	0.09	0.52

- **Revenue channel.** The lack of broad inpatient price increases and patient-demographic shifts weakens a pure price-extraction interpretation.
- **Institutional channel.** CPOM legal variation supports the claim that PE entry itself is associated with restructuring, not merely that already-restructuring hospitals attract PE.
- **Heterogeneity channel.** Nonprofit targets appear to offer greater administrative-efficiency gains, implying that organizational form matters for where PE can generate operating improvements.

Interpretation. The overall conclusion is conditional and multidimensional: PE hospitals become leaner and more profitable; the observable cost reductions are concentrated in overhead rather than long-run clinical staffing; patients report worse experiences; and hard clinical outcomes do not show measurable deterioration in the paper's samples.

7 Short summary

- PE-acquired hospitals maintain survival rates and improve short-run profitability.
- PE targets reduce total employment and wage expenditures.
- Core medical workers temporarily decline but recover; administrative workers remain lower.
- CPOM relaxation supports the PE mechanism by increasing PE acquisition likelihood.
- Previously nonprofit hospitals experience especially large administrative-worker cuts.
- Patient demographics and average inpatient prices do not significantly change.
- Patient satisfaction declines, but mortality and readmission rates do not deteriorate.
- The paper's evidence does not support the strongest anti-PE view that PE mainly closes hospitals, cuts clinical capacity permanently, raises inpatient prices broadly, or worsens measured mortality.
- The evidence also does not support a purely benign view: PE ownership has real labor-market consequences and reduces patient satisfaction across several dimensions.
- The most accurate summary is that PE changes hospital operations through administrative restructuring, producing financial gains with mixed patient-facing consequences.

8 Conclusion

8.1 Core conclusion

PE improves operating efficiency in acquired hospitals largely through administrative cost reductions, without evidence of broad inpatient price increases, patient-demographic changes, excess closures, or worse mortality/readmission outcomes. The clearest operational change is a durable reduction in administrative workers and administrative wage costs.

8.2 Economic intuition

PE owners appear to impose a sharper operating discipline on hospitals. The paper's results suggest that they reduce overhead and restructure hospital employment while avoiding a permanent reduction in core medical staffing. This can raise profitability without necessarily changing the patient mix or inpatient prices. However, administrative labor is not irrelevant to patients: fewer administrative resources can mean worse communication, less responsiveness, and weaker patient experience even when clinical outcomes remain stable.

8.3 What the paper shows

- PE targets remain open at rates comparable to matched controls.
- PE targets improve operating profitability in the short run.
- Employment and wage reductions are real and economically meaningful.
- The persistent labor reduction is concentrated in administrative workers, not long-run core medical workers.
- Nonprofit targets experience especially large administrative restructuring.
- Patient satisfaction declines across multiple HCAHPS dimensions.
- The paper does not detect worse mortality or readmission outcomes for PE targets.

8.4 What the paper does not show

- It does not show that PE ownership is costless for patients; patient satisfaction worsens.
- It does not show that PE improves clinical quality; it shows no detected deterioration in selected mortality and readmission outcomes.
- It does not rule out price effects in outpatient markets or other services outside the inpatient-price sample.
- It does not directly measure all post-buyout financial policies, such as leverage, dividend recapitalization, or financial fragility.
- It does not imply that every PE hospital deal is beneficial; the estimates describe average effects in the studied sample.

8.5 Policy and managerial implications

- For regulators, the paper suggests that monitoring should not focus only on closures or mortality. Patient experience, staffing composition, and administrative capacity also matter.
- For hospital managers, the results suggest that administrative efficiency can be improved, but excessive reductions may degrade patient-facing services.

- For investors, the findings indicate that PE value creation in hospitals can come from operational discipline rather than price increases, but reputational and regulatory risks may arise from lower satisfaction.
- For researchers, the paper motivates studying outpatient prices, post-buyout financial leverage, emergency care, staffing quality, and long-run access to care.

8.6 Final takeaway

The paper provides a nuanced view of private equity in hospitals. PE ownership is not portrayed as simply value destroying, because hospitals do not close at unusually high rates and hard clinical outcomes do not deteriorate in the observed measures. It is also not portrayed as purely efficiency enhancing, because labor cuts are substantial and patient satisfaction falls. The central lesson is that PE creates financial gains by making hospitals leaner, especially administratively, and that this leanness has both efficiency benefits and patient-experience costs.